

Practice problems #26

1. In $\mathbf{P}_2$, find the change-of-coordinates matrix from the basis $B = \{1 - 2t + t^2, 3 - 5t + 4t^2, 2t + 3t^2\}$ to the standard basis $S = \{1, t, t^2\}$.
If $p(t) = -1 + 2t$, what is the S -coordinate vector for $p(t)$?
Then find the B -coordinate vector for $p(t)$ *using the change-of-coordinates matrix*.
2. In $\mathbf{P}_1$, the following B and C are two bases. Find the change-of-coordinates matrix from B to C , and the change-of-coordinates matrix from C to B .

$$B = \{7 + 5t, -3 - t\}, \quad C = \{1 - 5t, -2 + 2t\}$$

Hints or Answers are on page 2.

#1: $M_{S \leftarrow B} = \begin{bmatrix} [b_1]_S & [b_2]_S & [b_3]_S \end{bmatrix} = \begin{bmatrix} 1 & 3 & 0 \\ -2 & -5 & 2 \\ 1 & 4 & 3 \end{bmatrix}$. What is $[p(t)]_S$?

$$[p(t)]_B = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

#2: Using isomorphism idea from $\mathbf{P}_1$ onto $\mathbf{R}^2$, that is, by using coordinate mapping relative to the standard basis $S = \{1, t\}$, it would be better to write the bases B and C as set of vectors in $\mathbf{R}^2$. Then, you can apply the algorithm that we did in class to find the change of coordinate matrix $M_{C \leftarrow B}$.

Ans:

$$M_{C \leftarrow B} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}, \quad \text{and} \quad M_{B \leftarrow C} = \begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}$$

Note that #1 is easier with definitions, #2 is easier with algorithms.